RULE SET INTERPRETABILITY WITH LARGE LANGUAGE

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INTRODUCTION

feedzai

Company specialized in financial fraud detection

PROBLEM

The detection of fraud is based on specific rule sets

Decisions must be explained to stakeholders

Complex rule sets are challenging to understand

GOAL

Leverage LLMs to generate clear descriptions and names for each rule

Make rules more interpretable for end users

Build trust in each decision

PROBLEM

DATA SET

Bank Account Fraud (BAF)

Fraud detection dataset

RIFF

606

Rule Generation

50

Rule: housing_status_BA > 0.5 ∧ proposed_credit_limit > 1700.0

MANUALLY GENERATED

Description: The rule is checking if the house status corresponds to “BA” and if the applicant’s proposed credit limit is high, with a value above 1700

Name: BA Housing with high credit limit

TASK

Train LLMs to generate descriptions and names. Measure performance with ROUGE1

RECALL

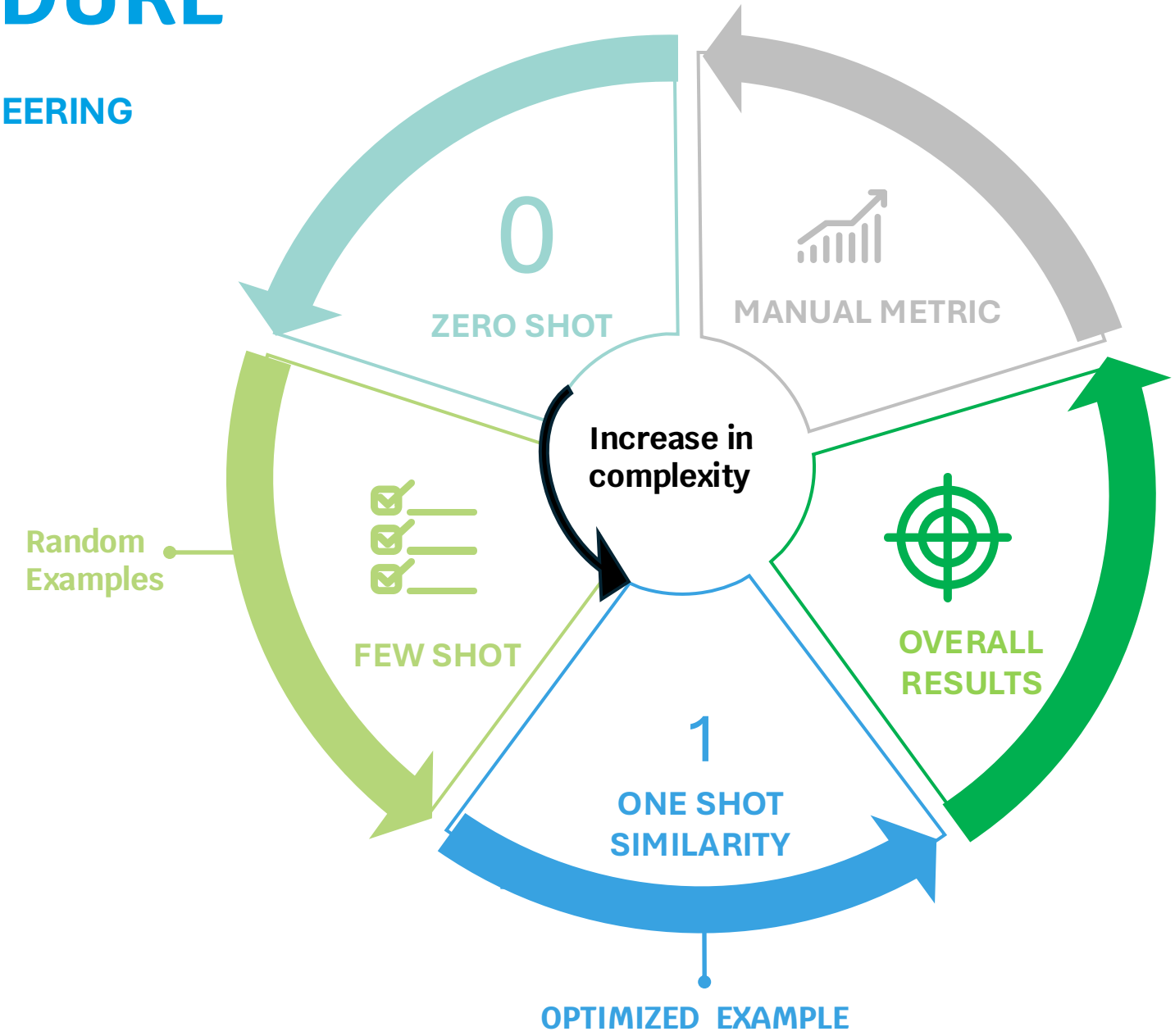
$$\frac{\text{\# overlaping unigrams}}{\text{total \# of unigrams in the reference}}$$

PRECISION

$$\frac{\text{\# overlaping unigrams}}{\text{total \# of unigrams in the output}}$$

PROCEDURE

PROMPT ENGINEERING



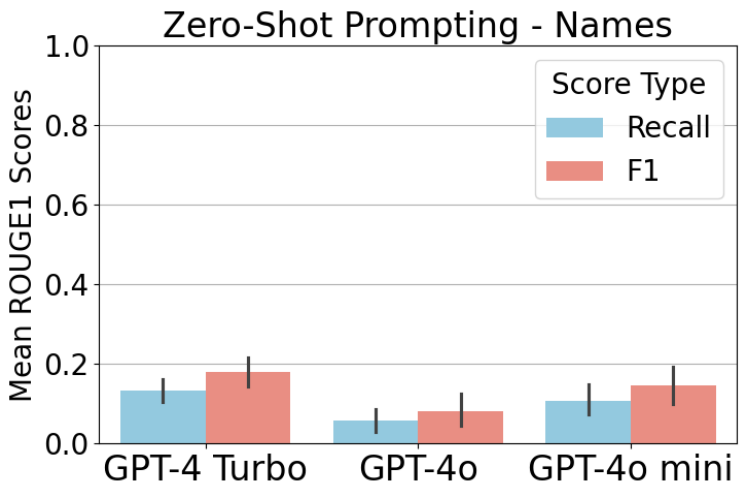
ZERO-SHOT PROMPTING

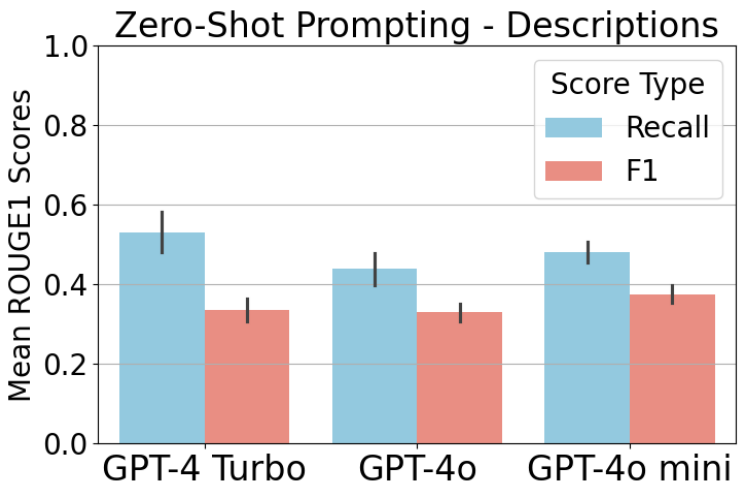
ZERO-SHOT PROMPTING

PROMPT

" Create a name and a description for this rule: < Test rule goes here >
Name:
Description:"

RESULTS





Names: depend heavily on key variables. More concise nature

Descriptions: Recall higher than F1

Overall results are not strong

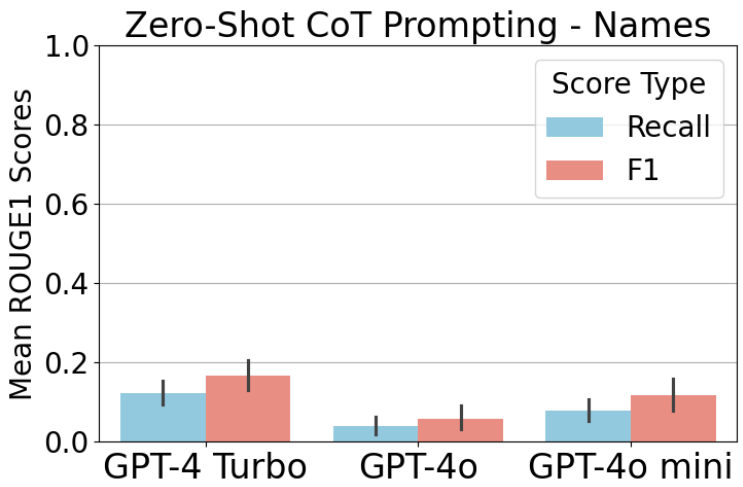
ZERO-SHOT PROMPTING

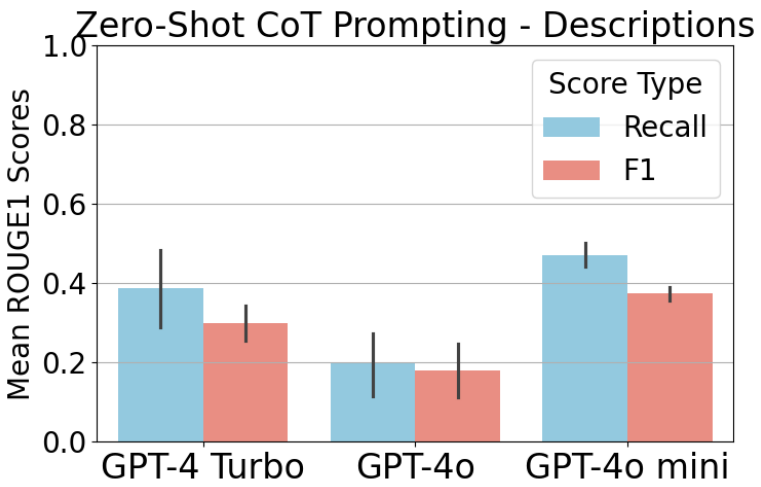
ZERO-SHOT PROMPTING COT

PROMPT

" Create a name and a description for this rule: < Test rule goes here >
Name:
Description:
Let's think step by step. "

RESULTS





General decline in the results

Not a completion task anymore

More commonly used for numerical problems

FEW-SHOT PROMPTING

BEST NUMBER OF SHOTS

NUMBER OF SHOTS

Gradually adding information to the prompt

Increasing number of shots

1-5 examples

BEST NUMBER OF SHOTS

TEMPERATURE

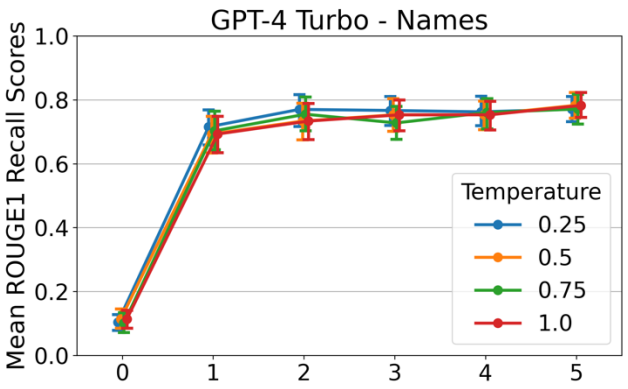
Influences randomness of output

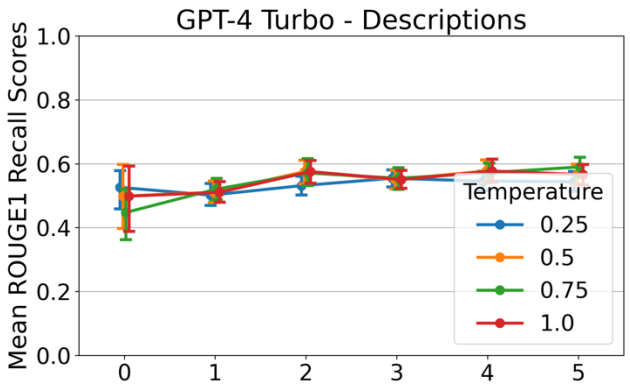
Lower temperature - more deterministic

Higher temperatures – greater randomness

BETTER TEMPERATURE

RESULTS





	SHOTS	TEMP
GPT-4 Turbo	**2**	**0.25**
GPT-4o	**2**	**1**
GPT-4o mini	**2**	**1**

FEW-SHOT PROMPTING

TWO-SHOT PROMPTING

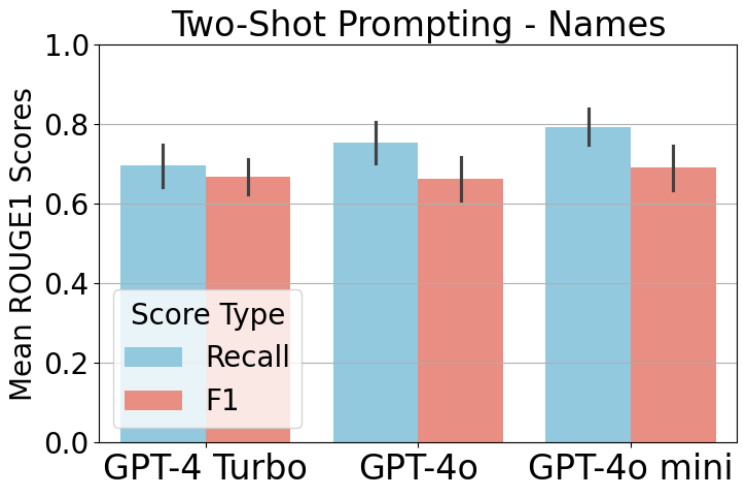
PROMPT

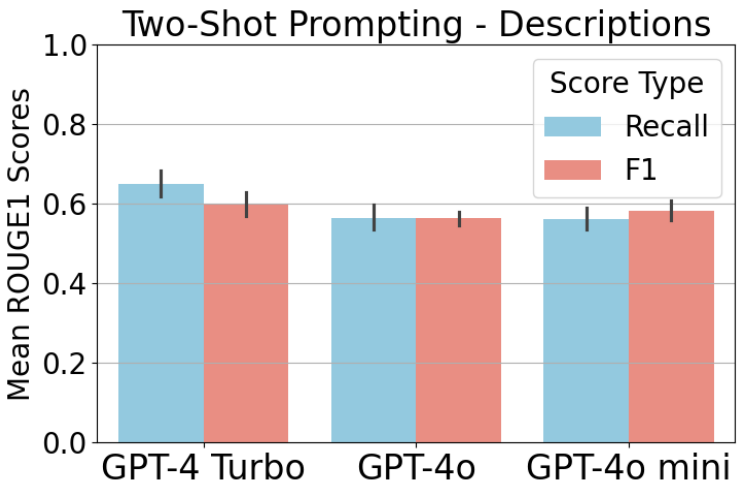
"**Rule**: < First rule goes here >
Description: < First description goes here >
Name: < First name goes here >

Rule: < Second rule goes here >
Description: < Second description goes here >
Name: < Second name goes here >

Rule: < Test rule goes here > "

RESULTS





Names: key variables are identified

Descriptions: examples offer clear guidance to the model

Greater increase for the names

FEW-SHOT PROMPTING

TWO-SHOT PROMPTING COT

PROMPT

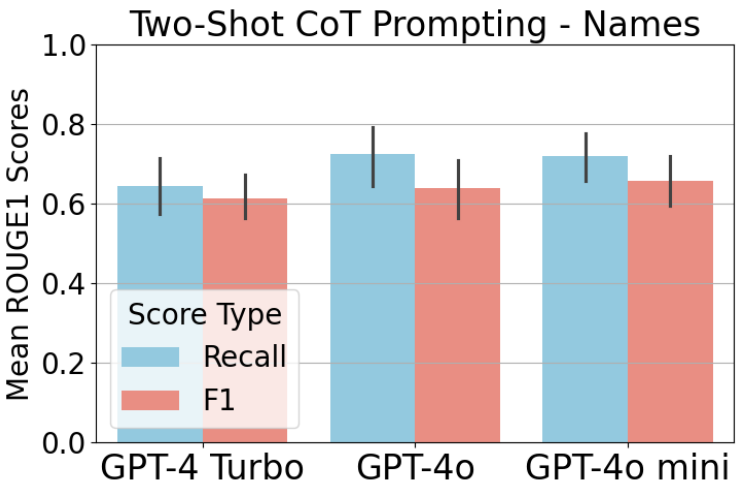
"Rule: < First rule goes here >
Description: < First description goes here >
Name: < First name goes here >

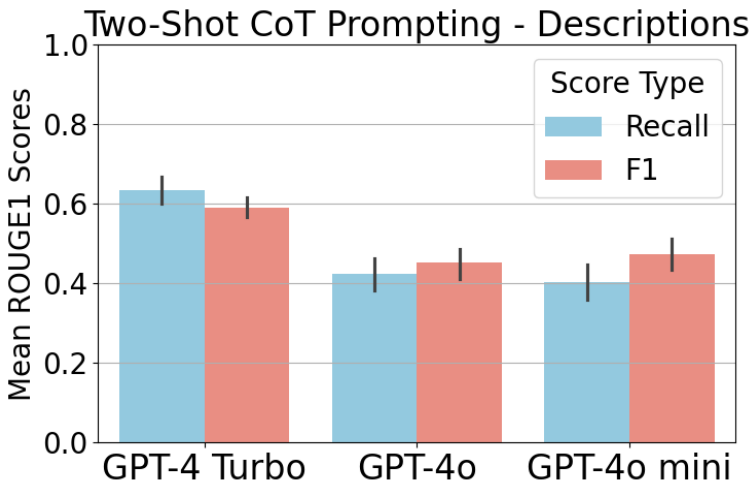
Rule: < Second rule goes here >
Description: < Second description goes here >
Name: < Second name goes here >

Rule: < Test rule goes here >

Let's think step by step. End the output with 'Description:' and 'Name:' sections"

RESULTS





Focus on explaining step by step, and not as much on the examples

General decline in the results

ONE-SHOT SIMILARITY

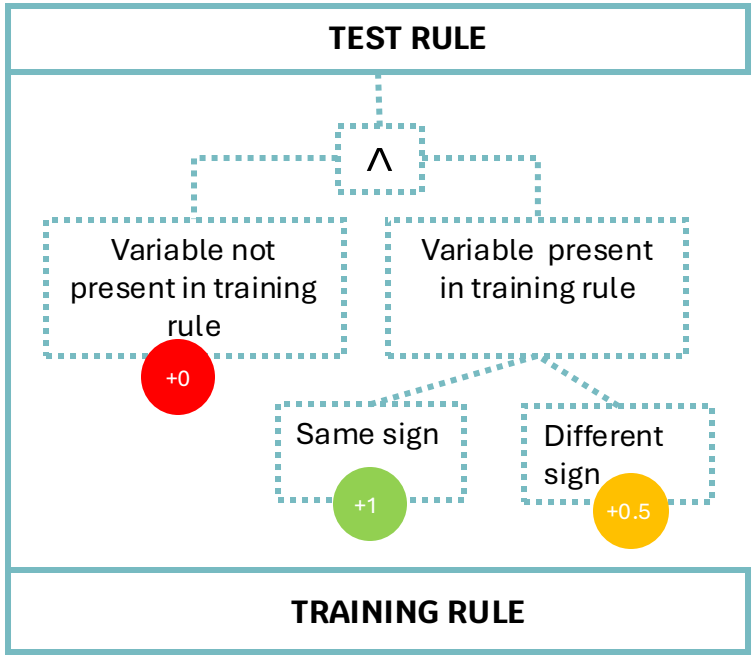
WHY ONE SHOT SIMILARITY?

1

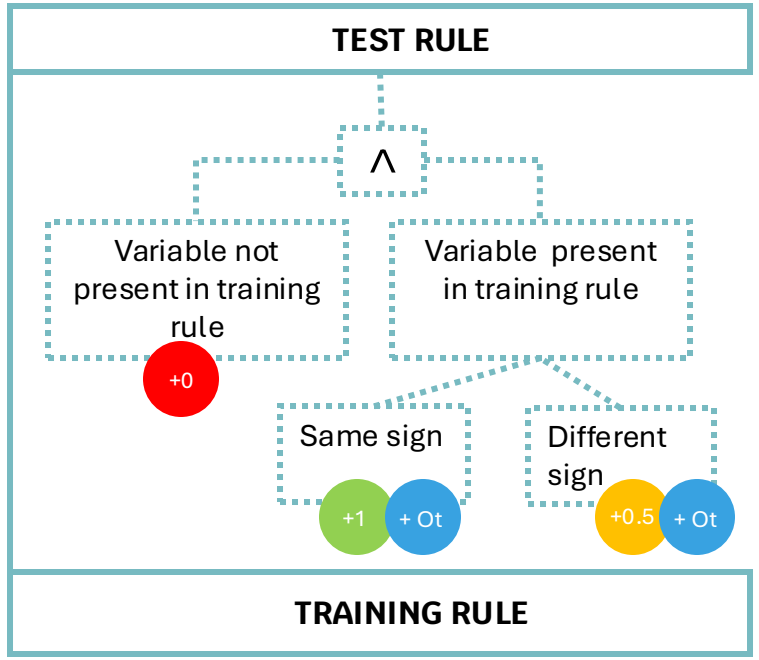
Improvement from **0** to **1** shot is significantly greater than improvement from **1** to **2** shots

3 DIFFERENT SIMILARITY MEASURES

OS SIM



OS SIM ORDER



$Ot = 1 / (1 + i)$

Ot: Order term
i : absolute diffence in position

OS SIM BERT

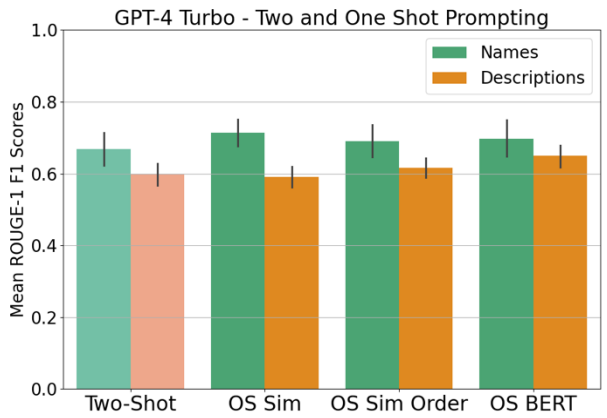
BERT

Embeddings compared using

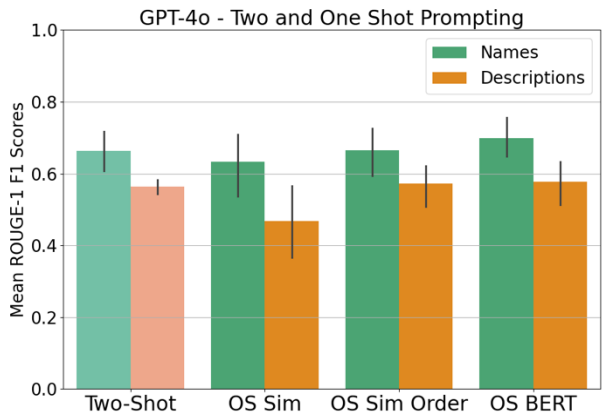
cosine similarity

ONE-SHOT SIMILARITY

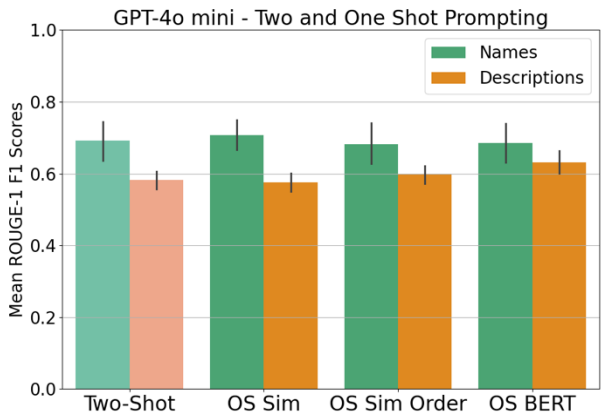
GTP-4 TURBO



GTP-4O



GTP-4O MINI



1 vs. 2

OS Sim: effective strategy for generating high-quality outputs.
Often outperforms the two-shot baseline.

Best Similarity

Three measures show strong performance across both names and descriptions, with variations in which approach performed best. Each method offers unique advantages in capturing nuanced patterns.

Main Conclusion

Targeted example selection using similarity measures offers significant advantages over relying on more but randomly chosen exemples.

MANUAL RULE CHECKING METRIC

APPROACH

HOW IT WORKS

Check if generated output descriptions **included all the variables** from the test rule and introduced any **extra** ones not present in the input but included in the dataset.

Manually defined key-words and rules associated with each variable.

customer_age:
"age" OR "years old" OR "twenties" OR "thirties" OR "forties" OR "fifties" OR "sixties" OR "seventies"

keep_alive_session:
"session" AND ("alive" OR "active" OR "running" OR "open" OR "uninterrupted" OR "ongoing" OR "sustained" OR "without interruption" OR "no interruption")

COMPARE TO ROUGE

RECALL:

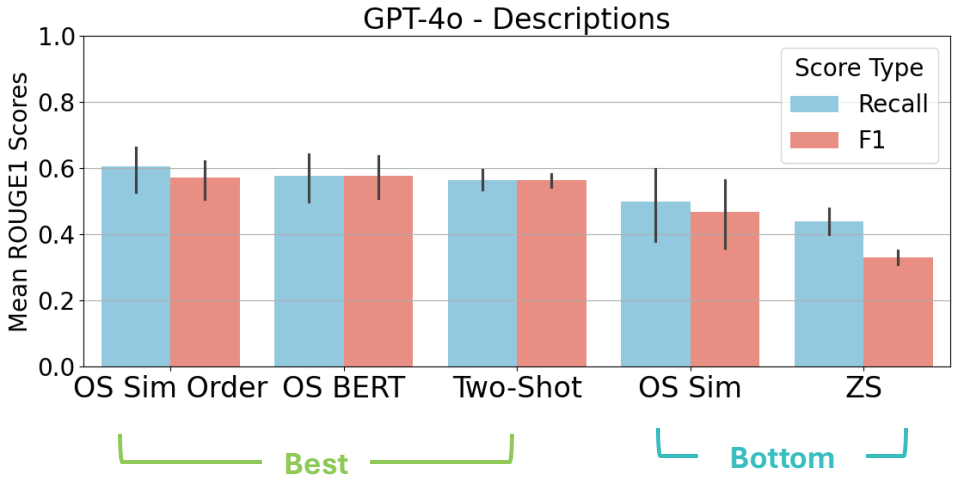
$$\frac{\text{\# correct variables in the description}}{\text{total \# of variables in the rule}}$$

PRECISION:

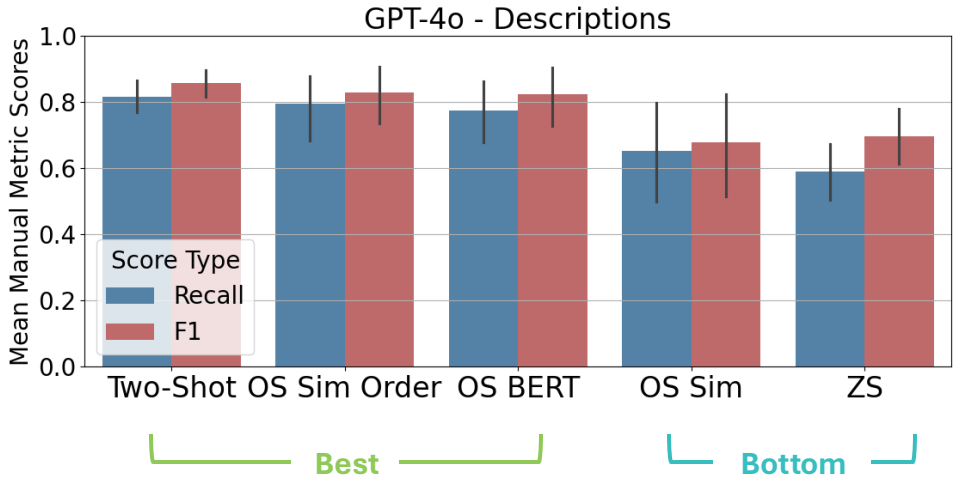
$$\frac{\text{\# correct variables in the description}}{\text{total \# of variables in the description}}$$

MANUAL RULE CHECKING METRIC

ROUGE-1



MANUAL RULE CHECKING METRIC



BEST

Two-shot: more random exemples → high chance of new variables → can't imitate directly
ROUGE fails to detect synonims → 3rd ROUGE to 1st Manual Metric
One-shot: most similar exemple → high chance of same variables → can imitate

BOTTOM

One-shot sim: lower performance than its counterparts
Zero-shot: lexical similarity may be lower but captures a comparable proportion of relevant variables

CONCLUSIONS

FRAUD DETECTION

Use of LLMs to generate names and descriptions for rules shows significant potential in improving the transparency and communication of those rules

PROMPTING STRATEGY

One-Shot prompting based on similarity, particularly considering the order or leveraging BERT, produces results at the level of multiple shot prompting or even better

Best results and reduced computational cost

EVALUATION METRICS

ROUGE-1 is valuable for measuring lexical overlap with reference

The Manual Rule Checking Metric revealed interesting information about the prompting techniques

Combining both analyses may offer a more holistic assessment